

## Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022 Semester 1

### (Lab 8)

(Practicing Lab, NOT for submission)

Review the related lecture note(s), and finish exercises below.

\* May use `main()` function approach (or multiple-functions) learnt in our earlier lesson in this lab

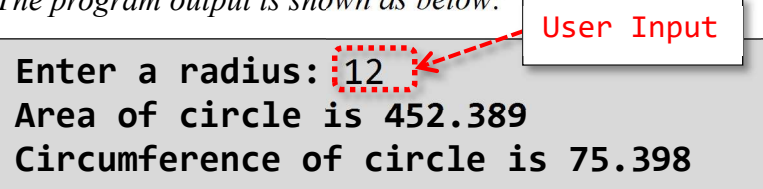
**Task:** build a program (for handling circles), with Procedural and Object-Oriented (OO) approaches:

- Given the radius of a circle, we need to calculate the *circumference* and *area* of the *circle*.
- Define a simple class of ICP students

1. **Procedural approach (multiple functions):** Write a Python program (named `Lab8_1.py`) with 3 functions:

- Function `main()` to 1) let user input the radius of a circle, and display the result of the circumference and area of the circle by properly calling the other two functions
  - Call the `main()` function to start execution as usual
- Function `circumference()`, which accepts a parameter of circle radius, calculate and returns the result value of the circle circumference
- Function `area()`, which accepts a parameter of circle radius, calculate and returns the result value of the circle area.

The program output is shown as below:



User Input

Enter a radius: 12  
Area of circle is 452.389  
Circumference of circle is 75.398

Sample codes for reference:

```
# Lab8_1.py, ICP, 2021
# Handling circles, using procedural approach (functions)

import math # import the math module

def circumference(radius): # a function
    return 2*math.pi*radius

def area(radius): # a function to find area
    return math.pi*radius*radius

def main(): # the main function
    radius = float(input("Enter a radius: "))
    print(f"Area of circle is {area(radius):.3f}")
    print(f"Circumference of circle is {circumference(radius):.3f}")

main() # start execution
```

2. **Object-Oriented approach (Class and Objects):** Write a Python program (named **Lab8\_2.py**) with a class of Circle which has:
- A field representing the radius of a circle object: radius
  - A constructor (initializer) `__init__()`, which accepts a parameter of circle radius as its field value
  - Method `circumference()`, calculate and returns the result value of the circle circumference
  - Method `area()`, calculate and returns the result value of the circle area.

*Sample Program Output:*

```
Enter a radius of Circle Object: 12
Area of circle is 452.389
Circumference of circle is 75.398
```

*Sample codes for reference:*

```
# Lab8_2.py, ICP, 2021
# Handling circles, using OO approach (& Class and Objects)

import math # import the math module

class Circle: # define the class of Circle
    def __init__(self, inRadius=0.0): # constructor
        self.radius = inRadius # a field / instance variable

    def circumference(self): # a method
        return 2*math.pi*self.radius

    def area(self): # a method to find area
        return math.pi*self.radius*self.radius

def main(): # the main function
    radius = float(input("Enter a radius of Circle Object: "))
    aCircle = Circle(radius) # create/instantiate a new object
    print(f"Area of circle is {aCircle.area():.3f}")
    print(f"Circumference of circle is {aCircle.circumference():.3f}")

main() # start execution
```

3. Write a Python program (named **Lab8\_3.py**) to define a class named `ICPStudent`, which has:
- Fields: student ID, student name, a list of 4 icp assignments marks
  - A constructor (initializer) `__init__()`, which accepts parameters of student ID, student name, and icp mark list (with a default list of all zero marks in 4 icp assignments)
  - Method `avgMark()`, to calculate and returns the average of all 4 icp marks
  - Method `__str__()`, to print the student information as the sample below with `print()`.

Add a function `main()`, to create two students (yourself and your classmate). Also print student's information with `print()` function, and function `__str__()`.

\* Hint: student may refer and re-use the sample `Student` class in lecture notes.

|                                                                             |
|-----------------------------------------------------------------------------|
| <b>ICPStudent</b>                                                           |
| <b>sID</b><br><b>sName</b><br><b>sMarks</b>                                 |
| <code>__init__()</code><br><code>avgMark()</code><br><code>__str__()</code> |

*Partial Sample Source Code*

```
# Lab8_3.py, ICP, 2021
# Define ICPStudent Class, using OO approach (& Class and Objects)

class ICPStudent: # define ICPStudent class
    def __init__(self, sid, sname, smarks=[0]*4): # constructor
# .. and MORE

def main(): # the main function
    me = ICPStudent(20214020, "CHAN Siu Ming", [100,90,99,100])
    myFriend = ICPStudent(20214321, "CHAN Tai Man")
    print(me)
    print(myFriend)

main() # start execution
```

*Sample Program Output:*

```
SID:20214020, Name:CHAN Siu Ming, average ICP mark:97.25
SID:20214321, Name:CHAN Tai Man, average ICP mark:0.0
```